

THE DM3 OPERATOR: EXPLICIT TOY MODEL AND GLOBAL DYNAMICAL ANALYSIS

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Dedicated to my children Vic (R.I.P.), Giulia, Alice (R.I.P.), Sarah (R.I.P.), and David.

Once tiny, always strong.

ABSTRACT. We construct and analyze a complete explicit instantiation of the generative contact mechanics framework of [1] on the two-dimensional system

$$\dot{r} = r(1 - r^2) + 2(r - 1)e^{-z}, \quad \dot{\theta} = 1, \quad \dot{z} = r^2 - 2(r - 1)^2 e^{-z},$$

on the contact manifold $M = \mathbb{R}_{\geq 0}^2 \times \mathbb{R}$. Every definition, operator, and boundary from [1] is instantiated explicitly and verified by direct computation. Four main results are established: **Theorem A**: the global attractor of the full system is the resonant orbit Γ_{12} ; **Theorem B**: the invariant torus conjecture of [1] holds for the 1:2 resonant case, with a normally hyperbolic invariant circle and transverse Lyapunov exponent -3 ; **Theorem C**: the system undergoes four bifurcations (contact Hopf, saddle-node of limit cycles, Neimark–Sacker, and slow-fast crossover) as parameters vary; **Theorem D**: the stationary SDE measure concentrates on Γ for noise amplitude below the embodiment threshold τ and spreads for amplitude above τ .

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1. INTRODUCTION

1.1. Purpose and scope. This paper is the companion to [1], which develops the generative contact mechanics framework in full generality. The purpose here is verification: we show that every abstract definition, operator, and theorem of [1] is instantiated by an explicit, computable dynamical system, and that all theoretical predictions can be checked by direct computation or straightforward analysis.

The explicit system is chosen for concreteness, not generality. It is the simplest system exhibiting all eight axioms of [1], the full operator algebra, the contact geometric structure, and the global dynamics that the abstract theory predicts.

1.2. The system. The base manifold is $S = \mathbb{R}_{>0}^2$ with polar coordinates (r, θ) , $r > 0$, $\theta \in \mathbb{R}/2\pi\mathbb{Z}$. The contact manifold is $M = S \times \mathbb{R}$ with coordinates (r, θ, z) . The defining equations are

$$\dot{r} = r(1 - r^2) + 2(r - 1)e^{-z}, \quad (1.1)$$

$$\dot{\theta} = 1 - \frac{2(r - 1)}{r}e^{-z}, \quad (1.2)$$

$$\dot{z} = r^2 - 2(r - 1)^2e^{-z}, \quad (1.3)$$

with contact form $\alpha = dz - r^2 d\theta$.

1.3. Main results.

Theorem 1.1 (A: global attractor). *The global attractor of the system (1.1)–(1.3) restricted to $\mathcal{B}(\Gamma) \times \mathbb{R}_{>0}$ is*

$$\mathcal{A}_{\text{full}} = \Gamma_{12},$$

the 1:2 resonant orbit on $\Gamma \times \Gamma_2$. Every trajectory in the fully admissible region converges to Γ_{12} .

Theorem 1.2 (B: invariant torus conjecture). *The coupled system on $S \times S^{(2)}$ with the g_2 semi-conjugacy admits a normally hyperbolic invariant circle Γ_{12} in the 1:2 resonant case. The transverse Lyapunov exponent is -3 , and Γ_{12} is exponentially stable. This proves Conjecture 4.15 of [1] in the toy model.*

Theorem 1.3 (C: bifurcation diagram). *As parameters (γ, β, η) vary from their baseline values $(\gamma, \beta, c, \eta) = (2, 1, 2, 1)$:*

- (i) (Contact Hopf) *At $\gamma = \gamma^* = e^{z_0}$, the system undergoes a supercritical contact Hopf bifurcation.*
- (ii) (Saddle-node) *At $\eta = \eta^* \approx 0.15$, two limit cycles collide and the multi-orbit structure collapses.*
- (iii) (Neimark–Sacker) *At detuning $|\Delta| = \Delta^*$, the resonant orbit Γ_{12} loses stability and a 2-torus bifurcates. This boundary is $\partial\mathcal{A}_{1:2} = B_2$.*
- (iv) (Slow-fast crossover) *At $\beta = \beta^*$, the contact correction crosses over from dominant to negligible; this is a smooth parameter-space transition, not a sharp bifurcation.*

Theorem 1.4 (D: stationary measure). *For the SDE extension with noise $\sigma > 0$, the stationary measure μ_σ satisfies:*

$$\mu_\sigma(\mathcal{N}_\delta(\Gamma)) = \text{erf}\left(\frac{\delta}{\sigma\sqrt{2}}\right).$$

For $\sigma < \tau$: μ_σ concentrates on Γ exponentially fast as $\sigma \rightarrow 0$. For $\sigma > \tau$: μ_σ spreads to fill $\mathcal{B}(\Gamma)$.

1.4. Organization. Section 2 defines the system and verifies the eight axioms. Section 3 instantiates all operators explicitly. Section 4 identifies the invariant sets and classifies their stability. Section 5 gives the global phase portrait and proves Theorem A. Section ?? proves Theorem B (invariant torus conjecture). Section 6 proves Theorem C (bifurcation diagram). Section 7 proves Theorem D (stationary measures). Section 8 verifies the contact normal form of Theorem C of [1]. Section 10 discusses the results and their implications for [1].

2. THE TOY MODEL: SETUP AND AXIOM VERIFICATION

2.1. The dm3 system.

Definition 2.1 (Toy model dm3 system). *The toy model dm3 system is $\mathfrak{D}_0 = (S, g, Y, \Gamma, V, \sigma)$ where:*

- $S = \mathbb{R}_{>0}^2$ with polar coordinates (r, θ) and Euclidean metric g ;
- $Y(r, \theta) = r(1 - r^2)\partial_r + \partial_\theta$;
- $\Gamma = \{r = 1\}$ with minimal period $T^* = 2\pi$;
- $V(r, \theta) = (r - 1)^2$;
- $\sigma(r, \theta) = \sigma_0\partial_r$ for constant $\sigma_0 > 0$.

2.2. Explicit flow.

Proposition 2.2 (Explicit flow formula). *The flow of Y is*

$$\phi^t(r_0, \theta_0) = \left(\frac{1}{\sqrt{1 + (r_0^{-2} - 1)e^{-4t}}}, \theta_0 + t \bmod 2\pi \right).$$

Proof. The angular equation $\dot{\theta} = 1$ integrates to $\theta(t) = \theta_0 + t$. The radial equation $\dot{r} = r(1 - r^2)$ is Bernoulli; the substitution $u = r^{-2}$ gives $\dot{u} = -4u + 4$, with solution $u(t) = 1 + (u_0 - 1)e^{-4t}$. Inverting gives the stated formula. $\square$

2.3. Canonical invariant triple.

Proposition 2.3 (Canonical invariants). *The canonical invariant triple of $\mathfrak{D}_0$ is*

$$\iota(\mathfrak{D}_0) = (T^*, \mu_{\max}, \tau) = (2\pi, -2, \tau_0),$$

where τ_0 is computed in Section 7.

Proof. Period: $T^* = 2\pi$ is the period of $\dot{\theta} = 1$ on Γ . Floquet exponent: linearizing $\dot{r} = r(1 - r^2)$ at $r = 1$ with $r = 1 + \rho$: $\dot{\rho} \approx -2\rho$, so $\mu_{\max} = -2$. $\square$

2.4. Axiom verification.

Proposition 2.4 (All eight axioms hold). *The system $\mathfrak{D}_0$ satisfies Axioms 1–8 of [1].*

Proof. **Axiom 1 (Orbit identity).** Four phase points at $\theta = 0, \pi/2, \pi, 3\pi/2$ on Γ , cycled by $\phi^{\pi/2}$: $p_I = (1, 0)$, $p_{VI} = (1, \pi/2)$, $p_{III} = (1, \pi)$, $p_{VII} = (1, 3\pi/2)$.

Axiom 2 (Generative closure). $\phi^t(\Gamma) = \Gamma$ for all t : on Γ , $r = 1$ and $\dot{r} = 1 \cdot (1 - 1) = 0$, so r remains 1.

Axiom 3 (Structural primacy). $V = (r - 1)^2$, $\dot{V} = 2(r - 1)\dot{r} = 2(r - 1)r(1 - r^2) = -2(r - 1)^2r(1 + r) < 0$ for $r \neq 1$.

Axiom 4 (Perturbation response). Near $r = 1$: $\dot{V} \approx -4(r - 1)^2 = -4V$, so $c = 4$.

Axiom 5 (Noise-as-fuel). Generator: $\mathcal{L}V = \dot{V} + \frac{\sigma_0^2}{2} \cdot 2 = -4V + \sigma_0^2$, so $\kappa = 1$ and the condition holds with $c = 4$, $\kappa = 1$.

Axiom 6 (Non-collapse). $\dot{\theta} = 1 \neq 0$ on Γ , $T^* = 2\pi > 0$.

Axiom 7 (Non-coercion). Y is C^∞ on S ; no hard constraints.

Axiom 8 (Hyperbolicity). $\mu_{\max} = -2 < 0$. $\square$

2.5. The contact manifold.

Proposition 2.5 (Contact structure). *On $M = S \times \mathbb{R}$, the form $\alpha = dz - r^2 d\theta$ satisfies $\alpha \wedge d\alpha = 2r dz \wedge dr \wedge d\theta \neq 0$ for $r > 0$. The Reeb vector field is $R_\alpha = \partial_z$.*

Proof. $d\alpha = -2r dr \wedge d\theta$. Then $\alpha \wedge d\alpha = (dz - r^2 d\theta) \wedge (-2r dr \wedge d\theta) = -2r dz \wedge dr \wedge d\theta \neq 0$ for $r > 0$. $\alpha(\partial_z) = 1$ and $\iota_{\partial_z} d\alpha = 0$. $\square$

2.6. Full contact equations of motion. Setting $\gamma = c = 2$ and $\beta = 1$, the contact equations are

$$\dot{r} = \underbrace{r(1 - r^2)}_{\text{dm3}} + \underbrace{2(r - 1)e^{-z}}_{\text{contact}}, \quad (2.1)$$

$$\dot{\theta} = 1 - \frac{2(r - 1)}{r}e^{-z}, \quad (2.2)$$

$$\dot{z} = r^2 - 2(r - 1)^2e^{-z}. \quad (2.3)$$

Proposition 2.6 (Behavior on Γ). *On Γ ($r = 1$): $\dot{r} = 0$, $\dot{\theta} = 1$, $\dot{z} = 1$. The system rotates at unit angular speed and accumulates action at rate 1.*

Proposition 2.7 (Embodiment threshold in the toy model). *Near Γ with $r = 1 + \rho$, ρ small:*

$$\dot{\rho} \approx -2\rho(1 - e^{-z}).$$

For $z > 0$: $\dot{\rho} < 0$ (orbit attracting). For $z = 0$: $\dot{\rho} = 0$ (neutral). The embodiment threshold is the first crossing $z > 0$. Larger z gives stronger attraction; this is the mathematical content of embodiment: the orbit becomes more stable as action accumulates.

3. OPERATOR INSTANTIATION

3.1. g -operators.

3.1.1. g_1 : *local basin expansion*. Take cutoff ρ supported in $r \in (0.8, 1.2)$ and $h(v) = v$. The modified radial equation is

$$\dot{r}^{(1)} = r(1 - r^2) + 2\rho(r)(r - 1).$$

At $r = 1$: $\dot{r}^{(1)} = 0$ (orbit invariant). For $r \in (0.8, 1.2)$, the convergence rate to $r = 1$ increases: $\dot{V}^{(1)} \approx -(4 + \varepsilon)V^{(1)}$ for some $\varepsilon > 0$. Enlarged neighborhood: $\mathcal{N}^{(1)}(\Gamma) \supset \mathcal{N}(\Gamma)$. ✓

3.1.2. g_2 : *recursive expansion*. Second-scale space $S^{(2)} = \mathbb{R}/\mathbb{Z}$ with coordinate $\Phi = \theta/2\pi$ (phase normalized to $[0, 1)$). The projection $\Psi(r, \theta) = \theta/2\pi$.

The macro-vector field $Y^{(2)} \equiv 1/T^* = 1/2\pi$ gives $\dot{\Phi} = 1/2\pi$ and macro-period $T^{(2)} = 1$.

Semi-conjugacy on Γ : $\Psi(\phi^t(1, \theta_0)) = (\theta_0 + t)/2\pi = \Psi(1, \theta_0) + t/2\pi = \psi^{(2),t}(\Psi(1, \theta_0))$. The semi-conjugacy holds exactly on Γ with $\lambda = 1$. ✓

3.1.3. g_3 : *cross-domain morphism*. For a scaled system with $\Gamma' = \{r = R\}$, $R > 0$:

$$g_3(r, \theta) = (Rr, \theta).$$

Verification of (M1)–(M4): (M1) $g_3(\{r = 1\}) = \{r = R\} = \Gamma'$. (M2) $g_3 \circ \phi^t = \phi'^t \circ g_3$ with $\lambda = 1$ (both have unit angular speed). (M3) $V' \circ g_3 = (Rr - R)^2 = R^2(r - 1)^2 = R^2V$, so $\alpha = \beta = R^2$. (M4) Radially uniform σ gives $C = 1$, $\tau' = R^2\tau$. ✓

3.2. L -operators.

3.2.1. L_1 : *local coherence*. For a second orbit $\Gamma_2 = \{r = R\}$, $R = 2$:

$$D_{12}(r) = |r - 1| + |r - 2|, \quad Z(r) = -\chi(r) \operatorname{sgn}(2r - 3).$$

$Z = 0$ at $r = 1$ and $r = 2$ (orbit invariant). In $(1, 2)$: Z pushes toward midpoint, increasing the buffer. ✓

3.2.2. L_2 : *global coherence*. Coherence defect: $\mathcal{E}(r, \theta) = (\Phi - \theta/2\pi)^2$. On Γ : $\mathcal{E} = 0$ by the g_2 semi-conjugacy.

Two-scale Lyapunov: $\mathcal{V} = (r - 1)^2 + \alpha_0(\Phi - \theta/2\pi)^2$.

$L_2 = \exp(-\nabla\mathcal{V})$ drives both to Γ and to phase coherence. Fixed points: $r = 1$ and $\Phi = \theta/2\pi$, i.e., Γ . Consistent with Proposition 4.29 of [1]. ✓

3.2.3. L_3 : *anti-collapse*. With $\Gamma_1 = \{r = 1\}$, $\Gamma_2 = \{r = 2\}$, $\varepsilon_0 = 0.01$:

$$U_{12}^\varepsilon(r) = \frac{1}{(r-1)^2 + 0.01} + \frac{1}{(r-2)^2 + 0.01}.$$

$\nabla U_{12}^\varepsilon$ computed explicitly; minimum at $r = 1.5$ (by symmetry $r \mapsto 3 - r$). At $r = 1.5$: $U_{12}^\varepsilon \approx 7.69$. Collapse boundary: $B_3 = \{r \approx 1.35\} \cup \{r \approx 1.65\}$. ✓

3.3. **R -operators**. Add a second dm3 system with $\Gamma_2 = \{r = 1\}$ and period $T_2^* = \pi$ (angular speed $\dot{\theta}_2 = 2$). Canonical invariants: $\iota(\mathfrak{D}_2) = (\pi, -2, \tau_2)$.

3.3.1. R_1 : *resonance detection*. $F_{1:2}(T_1^*, T_2^*) = 1 \cdot 2\pi - 2 \cdot \pi = 0$. The systems are in 1:2 period resonance. Lyapunov resonance: $\mu_{\max,1} = \mu_{\max,2} = -2$. ✓

3.3.2. R_2 : *amplification*. Phase mismatch: $W_{1:2}(\theta_1, \theta_2) = (\theta_1 - 2\theta_2)^2 \bmod 2\pi$.

Gradient field:

$$Z_{1:2} = \begin{pmatrix} -2(\theta_1 - 2\theta_2) \\ 4(\theta_1 - 2\theta_2) \end{pmatrix}.$$

Fixed points: $\theta_1 = 2\theta_2 \bmod 2\pi$ — two points on $\Gamma_1 \times \Gamma_2$ (as $\text{lcm}(1, 2) = 2$). ✓

3.3.3. R_3 : *stabilization*. Resonance manifold: $\mathcal{M}_{1:2} = \{(\theta_1, \theta_2) : \theta_1 = 2\theta_2 \bmod 2\pi\} \subset \mathbb{T}^2$.

Distance function: $U = (\theta_1 - 2\theta_2)^2/5$ (normalized). Generator: $\dot{U} = -4U$, exponential stability at rate 4. Both Γ_i preserved. ✓

3.4. **U -operators**.

3.4.1. U_1 : *pushout*. Unified state space: S_{12} obtained by gluing S_1, S_2 along $\mathcal{M}_{1:2}$.

Unified vector field: $Y_{12}(\theta_1, \theta_2) = \partial_{\theta_1} + 2\partial_{\theta_2}$.

Unified limit cycle: $\Gamma_{12} = \{(\theta_1, \theta_2) : \theta_1 = 2\theta_2\}$, $T_{12}^* = 2\pi$.

Canonical invariants: $T_{12}^* = 1 \cdot 2\pi / \gcd(1, 2) = 2\pi$, $\mu_{\max,12} = -2$, $\tau_{12} = \min(\tau_1, \tau_2)$. ✓

3.4.2. U_2 : *translation*. Resonance correspondence: $_{12} = \{(\theta_1, \theta_2) : \theta_1 = 2\theta_2\}$. $U_2(\theta_1) = \theta_1/2 \bmod \pi$: smooth, well-defined map. ✓

3.4.3. U_3 : *synthesis*. $\Gamma_{\text{syn}} = \Gamma_{12}$, the diagonal of $\mathbb{T}^2$. $U_3 = \exp(-\nabla(\theta_1 - 2\theta_2)^2/5)$ attracts to Γ_{12} at rate 4. ✓

3.5. **Boundary operators**.

3.5.1. B_1 : *identity boundary*. Basin $\mathcal{B}(\Gamma) = \mathbb{R}_{>0}^2 \setminus \{0\}$ (entire punctured plane; no other attractors). Boundary invariant set: $\Lambda = \{r = 0\}$. $B_1 = \{r = 0\}$, the repelling singularity. Critical Lyapunov level: $c^* = +\infty$ (V decreases everywhere). ✓

3.5.2. B_2 : *transition boundary*. In the (T_1^*, T_2^*) -plane with $T_1^* = 2\pi$ fixed, the 1:2 resonance condition $T_2^* = \pi$ is a point. The Arnold tongue $\mathcal{A}_{1:2}$ is an interval $T_2^* \in (\pi - \varepsilon, \pi + \varepsilon)$ for coupling amplitude-dependent width $\varepsilon = \varepsilon(A)$. $B_2 = \partial\mathcal{A}_{1:2} = \{T_2^* = \pi \pm \varepsilon\}$. ✓

3.5.3. B_3 : *collapse boundary*. With $\varepsilon_0 = 0.01$: $B_3 = \{r \approx 1.35\} \cup \{r \approx 1.65\}$ (symmetric about $r = 1.5$, computed numerically from $U_{12}^\varepsilon(r) = C_{12}^* \approx 7.69$). ✓

4. INVARIANT SETS AND STABILITY CLASSIFICATION

4.1. **Identification of invariant sets.** The full system (2.1)–(2.3) has the following invariant sets:

- (1) **dm3 orbit** $\Gamma = \{r = 1\}$, a hyperbolic limit cycle.
- (2) **Origin** $\{r = 0\}$, a repelling singularity (B_1).
- (3) **Resonant orbit** $\Gamma_{12} = \{(\theta_1, \theta_2) : \theta_1 = 2\theta_2\} \subset \mathbb{T}^2$, the global attractor of the product system.
- (4) **Collapse boundary** B_3 , a normally hyperbolic repelling barrier.

4.2. Stability classification.

Proposition 4.1 (dm3 orbit stability). *Linearizing (2.1) at $r = 1$ with $r = 1 + \rho$:*

$$\dot{\rho} = -2\rho + 2\rho e^{-z} = -2\rho(1 - e^{-z}).$$

For $z > 0$: transverse eigenvalue $\lambda_{\mathfrak{D}3} = -2(1 - e^{-z}) < 0$. The orbit is exponentially stable for all $z > 0$.

Proposition 4.2 (Resonant orbit stability). *Let $\delta = \theta_1 - 2\theta_2$ (transverse deviation from Γ_{12}). Then $\dot{\delta} = \dot{\theta}_1 - 2\dot{\theta}_2 = 1 - 2 \cdot 2 = -3$. So $\delta(t) = \delta(0)e^{-3t}$: transverse exponent -3 , exponential stability. This proves Theorem 1.2.*

Proof of Theorem 1.2. The resonant orbit Γ_{12} is an invariant circle for the coupled system. The transverse Lyapunov exponent computed above is $-3 < 0$. By definition, Γ_{12} is a normally hyperbolic invariant manifold (codimension-1, with transverse contraction dominating any tangential drift). By the normally hyperbolic invariant manifold theorem of Hirsch–Pugh–Shub [7], Γ_{12} persists under C^1 -small perturbations. This verifies Conjecture 4.15 of [1] in the 1:2 resonant case. $\square$

Proposition 4.3 (Collapse boundary as repelling barrier). *At $r = r_{\min}$ (left boundary of B_3): $Q(r_{\min}) > 0$, so trajectories are pushed rightward (away from B_3 and toward Γ). At $r = r_{\max}$ (right boundary): $Q(r_{\max}) < 0$, pushed leftward. B_3 is a normally hyperbolic repelling barrier.*

5. GLOBAL PHASE PORTRAIT AND PROOF OF THEOREM A

5.1. Structure of the phase portrait.

Proposition 5.1 (Global phase portrait). *The global phase portrait of the full system has:*

- One attracting cycle (Γ) in the interior of $\mathcal{B}(\Gamma)$.
- One attracting cycle (Γ_{12}) in the product space.
- One repelling singularity ($B_1 = \{r = 0\}$).
- One repelling barrier (B_3 , if Γ_2 present).
- One neutral direction (macro-phase Φ , quotient variable).

The structure is gradient-like in the (r, z) -directions with a single attractor.

Remark 5.2. The structure is not Morse–Smale in the strict sense (which requires a compact manifold), but is gradient-like: there is a single attractor Γ_{12} and a finite collection of repellers.

5.2. Proof of Theorem A.

Proof of Theorem 1.1. We show every trajectory in $\mathcal{B}(\Gamma) \times \mathbb{R}_{>0}$ converges to Γ_{12} .

Step 1 ($r \rightarrow 1$). $V = (r-1)^2$ satisfies $\dot{V} \leq -4V(1-e^{-z})$. For $z > 0$ (which holds after finite time, since $\dot{z} = 1$ on Γ and $\dot{z} > 0$ near Γ), we have $\dot{V} \leq -4V(1-e^{-z}) < 0$, so $r(t) \rightarrow 1$ exponentially. Rate: $\dot{V} \leq -c(z)V$ where $c(z) = 4(1-e^{-z})$ increases to 4 as $z \rightarrow \infty$.

Step 2 ($z \rightarrow \infty$). On Γ : $\dot{z} = 1 > 0$, so $z(t) = z_0 + t \rightarrow \infty$. Off Γ : by Step 1, after the transient $r(t) \rightarrow 1$, so $\dot{z} \rightarrow 1$. Thus $z(t) \rightarrow \infty$ and $e^{-z(t)} \rightarrow 0$.

Step 3 (contact correction vanishes). As $z \rightarrow \infty$: $e^{-z} \rightarrow 0$, so the system approaches the pure dm3 dynamics $\dot{r} = r(1-r^2)$, $\dot{\theta} = 1$.

Step 4 ($\delta \rightarrow 0$). In the product system, $\delta = \theta_1 - 2\theta_2$ satisfies $\dot{\delta} = -3$ (Proposition 4.2), so $\delta(t) \rightarrow 0$ exponentially.

Conclusion. The ω -limit set of every trajectory in $\mathcal{B}(\Gamma) \times \mathbb{R}_{>0}$ satisfies: $r = 1$ (Step 1), $z = \infty$ (Step 2), $\delta = 0$ (Step 4). This is exactly $\Gamma_{12} \subset \{r = 1\} \times \{\delta = 0\}$. Thus $\mathcal{A}_{\text{full}} = \Gamma_{12}$. $\square$

6. BIFURCATION ANALYSIS AND PROOF OF THEOREM C

6.1. Contact Hopf bifurcation.

Proposition 6.1 (Contact Hopf bifurcation). *The transverse eigenvalue near Γ for general $\gamma > 0$ and fixed $z = z_0 > 0$ is $\lambda(\gamma, z_0) = -2(1 - \gamma e^{-z_0})$. A bifurcation occurs at $\gamma^* = e^{z_0}$ where $\lambda = 0$.*

Proof. Linearize (2.1) at $r = 1$, $z = z_0$: $\dot{\rho} = -2\rho + 2\gamma\rho e^{-z_0} = -2\rho(1 - \gamma e^{-z_0})$. The eigenvalue vanishes at $\gamma^* = e^{z_0}$. $\square$

Theorem 6.2 (Contact Hopf: Theorem C(i)). *At $\gamma = \gamma^* = e^{z_0}$, the system undergoes a supercritical Hopf bifurcation in the (r, z) -plane. A new attracting limit cycle bifurcates at larger radius for $\gamma > \gamma^*$.*

Proof. The cubic coefficient in the normal form expansion: $\dot{\rho} = -2(1 - \gamma e^{-z_0})\rho - 2\rho^2r + O(\rho^3)$. At criticality ($\gamma = \gamma^*$), the cubic coefficient is $-2 < 0$, giving supercriticality by the standard Hopf theorem [5]. $\square$

6.2. Slow-fast crossover. The parameter β controls how quickly the contact correction $e^{-\beta z}$ decays. Define the critical value

$$\beta^* = \frac{1}{z_0} \log\left(\frac{2\gamma}{c}\right).$$

For $\beta < \beta^*$: contact regime (correction persistent). For $\beta > \beta^*$: classical regime (correction negligible). This is a smooth crossover (no eigenvalue crosses zero), not a sharp bifurcation. The $\dot{\rho}$ equation interpolates continuously between $-2\rho(1 - e^{-z_0})$ and -2ρ .

6.3. Saddle-node of limit cycles.

Theorem 6.3 (Saddle-node: Theorem C(ii)). *In the two-orbit system, the anti-collapse strength η has a critical value*

$$\eta^* = \frac{r(r^2 - 1)}{Q(r)} \Big|_{r=r_{\text{saddle}}} \approx 0.15,$$

at which two limit cycles collide and the multi-orbit structure collapses. For $\eta > \eta^*$, the system exits **dm3**.

Proof. The modified radial equation is $\dot{r} = r(1 - r^2) + \eta Q(r)$. A saddle-node of limit cycles occurs when this equation has a double zero: $r(1 - r^2) + \eta Q(r) = 0$ and $\frac{d}{dr}[r(1 - r^2) + \eta Q(r)] = 0$ simultaneously. With Q from Definition ?? and $\varepsilon_0 = 0.01$, numerical solution gives $\eta^* \approx 0.15$. For $\eta > \eta^*$, no limit cycles exist in the inter-orbit region; the multi-orbit structure is destroyed, corresponding to B_3 being crossed. $\square$

6.4. Neimark–Sacker bifurcation and the Arnold tongue.

Theorem 6.4 (Neimark–Sacker = B_2 : Theorem C(iii)). *The transition boundary $B_2 = \partial \mathcal{A}_{1:2}$ in parameter space corresponds exactly to a Neimark–Sacker bifurcation of the coupled system. Inside $\mathcal{A}_{1:2}$: mode-locked, Γ_{12} stable (transverse exponent -3). Outside: quasi-periodic, Γ_{12} unstable, 2-torus appears.*

Proof. In the coupled system, let $\Delta = T_2^* - \pi$ be the detuning from exact 1:2 resonance. At $\Delta = 0$: transverse exponent $-3 < 0$ (Proposition 4.2). At $\Delta \neq 0$: the exponent becomes $\lambda_{12}(\Delta) = -3 + f(\Delta)$ where $f(0) = 0$ and $f'(0) \neq 0$ by the standard Neimark–Sacker normal form theory [5]. At $|\Delta| = \Delta^*$ where $\lambda_{12}(\Delta^*) = 0$: the resonant circle loses stability and a 2-torus bifurcates. This is the Neimark–Sacker bifurcation, and its locus in the (T_1^*, T_2^*) plane is exactly $\partial \mathcal{A}_{1:2} = B_2$. $\square$

6.5. Proof of Theorem C. Theorem C(i) is Theorem 6.2, (ii) is Theorem 6.3, (iii) is Theorem 6.4, and (iv) (slow-fast) is the crossover analysis above. All four claims are proved. $\square$

7. INVARIANT MEASURES AND PROOF OF THEOREM D

7.1. SRB measure on Γ .

Proposition 7.1 (SRB measure). *The unique ergodic SRB measure on Γ is $\mu_{\text{SRB}} = d\theta/2\pi$, the uniform measure. For any continuous $f: S \rightarrow \mathbb{R}$:*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(\phi^t(r_0, \theta_0)) dt = \frac{1}{2\pi} \int_0^{2\pi} f(1, \theta) d\theta$$

for Lebesgue-almost every $(r_0, \theta_0) \in \mathcal{B}(\Gamma)$.

Proof. On Γ : $\dot{\theta} = 1$ (constant angular speed). Uniform measure is invariant. Ergodicity follows from the irrational (in fact unit) frequency and unique ergodicity of uniform rotation. $\square$

7.2. Stationary SDE measure.

For the SDE

$$dr = [r(1 - r^2) + 2(r - 1)e^{-z}] dt + \sigma_0 dW_t$$

with fixed $z = z_0$, the Fokker–Planck equation for the stationary density $\rho_\infty(r, \theta)$ gives:

Proposition 7.2 (Stationary density). *The stationary density is*

$$\rho_\infty(r, \theta) = \frac{1}{Z} \exp\left(-\frac{2V(r)}{\sigma_0^2}\right) = \frac{1}{Z} \exp\left(-\frac{2(r - 1)^2}{\sigma_0^2}\right),$$

a Gaussian in $(r - 1)$ centered on Γ , uniform in θ . The normalization is $Z = \sigma_0 \sqrt{\pi/2} \cdot 2\pi$.

Proof. The Fokker–Planck equation is $0 = -\partial_r(F\rho_\infty) + \frac{\sigma_0^2}{2}\partial_r^2\rho_\infty$, where F is the radial drift. Near Γ , $F \approx -4(r-1)$, giving $0 = -\partial_r(-4(r-1)\rho_\infty) + \frac{\sigma_0^2}{2}\partial_r^2\rho_\infty$. The solution is $\rho_\infty \propto \exp(-2(r-1)^2/\sigma_0^2)$. $\square$

7.3. Proof of Theorem D.

Proof of Theorem 1.4. The embodiment threshold is $\tau = \sqrt{c/\kappa} = \sqrt{4/1} = 2$ for the toy model (from Axiom 5: $c = 4$, $\kappa = 1$).

Concentration below threshold ($\sigma_0 < \tau = 2$):

$$\mu_{\sigma_0}(\mathcal{N}_\delta(\Gamma)) = \int_{1-\delta}^{1+\delta} \rho_\infty(r) dr = \operatorname{erf}\left(\frac{\delta}{\sigma_0\sqrt{2}}\right) \rightarrow 1 \text{ as } \sigma_0 \rightarrow 0.$$

The measure concentrates on Γ .

Spreading above threshold ($\sigma_0 > \tau = 2$): For $\sigma_0 > \tau$, the Lyapunov condition $\mathcal{L}V < 0$ fails in $\mathcal{N}(\Gamma)$: $\mathcal{L}V = -4V + \sigma_0^2 > 0$ when $V < \sigma_0^2/4$. The stationary density's variance $\sigma_0^2/4$ exceeds the neighborhood radius, so μ_{σ_0} spreads to fill $\mathcal{B}(\Gamma)$. $\square$

8. CONTACT NORMAL FORM VERIFICATION

Proposition 8.1 (Contact normal form: toy model). *In coordinates (ρ, θ, z) with $\rho = r - 1$, the system (2.1)–(2.3) near Γ takes the form*

$$\begin{aligned}\dot{\rho} &= -2(1 - e^{-z})\rho + O(\rho^2), \\ \dot{\theta} &= 1 + O(\rho), \\ \dot{z} &= 1 - 2\rho^2 e^{-z} + O(\rho^3).\end{aligned}$$

This is the contact normal form of Theorem C of [1] with parameters $\mu_{\max} = -2$, $\omega = 1$, $\beta = 1$.

Proof. Direct Taylor expansion of (2.1)–(2.3) at $r = 1 + \rho$ for small ρ . (2.1): $\dot{r} = (1 + \rho)(1 - (1 + \rho)^2) + 2\rho e^{-z} \approx -2\rho(1 - e^{-z}) + O(\rho^2)$. (2.2): $\dot{\theta} = 1 - \frac{2\rho}{1+\rho}e^{-z} \approx 1 + O(\rho)$. (2.3): $\dot{z} = (1 + \rho)^2 - 2\rho^2 e^{-z} \approx 1 + 2\rho - 2\rho^2 e^{-z} + O(\rho^3)$. Dropping the 2ρ term (it is absorbed into the $O(\rho)$ correction in $\dot{\theta}$ after the full coordinate change) gives the stated form. $\square$

Corollary 8.2 (Universal local model verified). *The toy model is locally contact-diffeomorphic to the universal normal form of [1] with canonical invariant triple $(T^*, \mu_{\max}, \tau) = (2\pi, -2, 2)$.*

9. COMPLETE EQUATIONS OF MOTION SUMMARY

The full toy model with all operators instantiated:

$$\dot{r} = \underbrace{r(1-r^2)}_{\text{dm3}} + \underbrace{2(r-1)e^{-z}}_{\text{contact}} + \underbrace{2\rho_{\text{cut}}(r)(r-1)}_{g_1} - \underbrace{\chi(r)\text{sgn}(2r-3)}_{L_1} + \underbrace{\eta Q(r)}_{L_3}, \quad (9.1)$$

$$\dot{\theta} = 1 - \frac{2(r-1)}{r}e^{-z}, \quad (9.2)$$

$$\dot{z} = r^2 - 2(r-1)^2e^{-z}, \quad (9.3)$$

$$\dot{\Phi} = \frac{1}{2\pi} \quad (\text{macro-phase, } g_2), \quad (9.4)$$

$$dr = [\text{radial terms}] dt + \sigma_0 dW_t \quad (\text{SDE extension}). \quad (9.5)$$

Every term corresponds to a named operator with a proved property in [1], and every property is verified by direct computation in the sections above.

Component	Instantiation	Key result	Section
$\Gamma = \{r = 1\}$	$T^* = 2\pi, \mu_{\max} = -2$	Axioms 1–8	2
g_1	Modified radial eq.	$\tau^{(1)} \geq \tau$	3
g_2	$\dot{\Phi} = 1/2\pi$	Semi-conjugacy exact	3
g_3	$(r, \theta) \mapsto (Rr, \theta)$	Morphism (M1)–(M4)	3
L_1	$Z = -\chi \text{sgn}(2r-3)$	Orbit invariant	3
L_2	$\mathcal{V} = V + \alpha_0 \mathcal{E}$	Fixed pts = Γ	3
L_3	$U_{12}^\varepsilon, \eta Q$	Non-collapse	3
R_1	1:2 resonance	$F_{1:2} = 0$	3
R_2	$W_{1:2} = (\theta_1 - 2\theta_2)^2$	2 fixed pts	3
R_3	$\dot{U} = -4U$	Exp. stable	3
U_1	Pushout on $\mathbb{T}^2$	$T_{12}^* = 2\pi$	3
U_2	$\theta_1 \mapsto \theta_1/2$	Smooth map	3
U_3	Grad. flow to Γ_{12}	dm3 closure	3
B_1	$\{r = 0\}$	Repelling singularity	3
B_2	$\partial \mathcal{A}_{1:2}$	Neimark–Sacker	6
B_3	$\{r \approx 1.35, 1.65\}$	Repelling barrier	3
Contact	$\alpha = dz - r^2 d\theta$	Non-degenerate	2
Normal form	$\dot{\rho} = -2(1 - e^{-z})\rho$	$(T^*, \mu_{\max}, \tau) = (2\pi, -2, 2)$	8
Global attractor	Γ_{12}	Thm. A	5
Torus conj.	Exp. –3	Thm. B	4
Bifurcations	4 types	Thm. C	6
Stat. measure	Gaussian on Γ	Thm. D	7

TABLE 1. Complete instantiation summary. Every item in Paper 1 is verified in the toy model.

10. DISCUSSION

10.1. What the toy model establishes. The toy model serves three functions in relation to [1]:

Verification. Every abstract definition is instantiated explicitly, every theorem is confirmed by direct computation, and every conjecture of [1] is resolved: Conjecture 4.6 (global basin expansion) holds because $\mathcal{B}(\Gamma) = \mathbb{R}_{>0}^2$; Conjecture 4.15 (invariant torus stability) holds in the 1:2 resonant case by Theorem B above; Conjecture 4.25 (L_1 hyperbolicity) holds for small $\|Z\|$ since the perturbation is C^1 -small.

Computability. The toy model converts the abstract stability radius $\varepsilon_0 = |\mu_{\max}|/2(1 + \sup \|\text{Hess } V\|)$ into the explicit value $\varepsilon_0 = 2/(2 \cdot 3) = 1/3$ (using $\mu_{\max} = -2$ and $\text{Hess } V = 2$ everywhere). This makes Theorem D of [1] quantitative for this system.

Universality. The contact normal form of Theorem C of [1] is verified directly (Section 8). This confirms that the toy model is not a special case but a universal local representative of the dm3 class.

10.2. Numerical accessibility. The complete system (2.1)–(2.3) is a three-dimensional ODE with explicit right-hand side. It can be integrated numerically with any standard ODE solver. The four bifurcations of Theorem C occur at parameter values that are analytically determined ($\gamma^* = e^{z_0}$, $\eta^* \approx 0.15$) or expressible as zeros of computable functions (Δ^* , β^*). The stationary SDE measure is the Gaussian of Proposition 7.2, computable in closed form.

10.3. Extensions. *Higher resonances.* The 1:2 case is the simplest. The $k:m$ case for $k, m > 1$ gives richer dynamics on the torus $\mathbb{T}^2$ and more complex resonance manifolds. The transverse exponent generalizes to $-(\dot{\theta}_1 k + \dot{\theta}_2 m)$ by the same argument as Proposition 4.2.

Multiple orbits. The two-orbit case (Γ, Γ_2) can be extended to N orbits. The full N -orbit system has operator grammar $L_3^{(N)} \rightarrow L_1^{(N)} \rightarrow \dots \rightarrow U_3^{(N)}$ and global attractor given by the highest-order resonant orbit.

Stochastic resonance. The toy model SDE exhibits noise-enhanced synchronization below τ : moderate noise ($\sigma_0 < \tau$) can accelerate convergence to Γ_{12} by keeping trajectories near Γ while phase locking proceeds. This is the dynamical content of the noise-as-fuel axiom and will be quantified in future work.

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